

Laboratory Record 3

Course :	Full Stack Web Development
Course Code:	23CM4121
Experiment no.:	3
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Roll No.:	A24126552108

1. TITLE

JavaScript Program using Arrays and Functions

2. AIM

To develop a JavaScript program using arrays and user-defined functions to calculate and display the total, average, highest, and lowest marks of students.

3. OBJECTIVE

- To understand the concept of arrays in JavaScript.
 - To learn how to create and use user-defined functions.
 - To calculate the total and average of marks using functions.
 - To find the highest and lowest marks using built-in JavaScript methods.
 - To use a for...of loop to process array elements.
 - To display the calculated results clearly using the console.
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4. THEORY

JavaScript Arrays are used to store multiple values in a single variable. In this program, an array named marks stores the marks of five students.

Functions are reusable blocks of code that perform specific tasks. The program uses separate functions to calculate the total, average, highest, and lowest marks.

The for...of loop is used to access each mark in the array and calculate the total. The average is calculated by dividing the total marks by the number of elements in the array using length.

JavaScript's Math.max() and Math.min() functions are used to find the highest and lowest marks. The spread operator (...) passes the array elements as individual values to these functions.

Finally, the console.log() method displays all calculated results in the browser or Node.js console.

5. ALGORITHM / PROCEDURE

1. Initialize an array marks with the marks of the students.
 2. Define a function calculateTotal() to calculate the sum of all marks using a for...of loop.
 3. Define a function calculateAverage() to calculate the average by dividing the total marks by the number of students.
 4. Define a function findHighest() using Math.max() and the spread operator to find the highest mark.
 5. Define a function findLowest() using Math.min() and the spread operator to find the lowest mark.
 6. Define a displayResult() function to display the marks and calculated results using console.log().
 7. Format the average to two decimal places using toFixed(2).
 8. Call the displayResult() function.
 9. Execute the program and verify the results in the console.
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6. PROGRAM

A : script.js

```
// Experiment 3: JavaScript Program using Arrays and Functions
// Student: R. Srivalli

// Array containing marks of students
let marks = [78, 85, 92, 67, 88];

// Function to calculate the total marks
function calculateTotal(values) {
    let total = 0;

    for (let mark of values) {
        total += mark;
    }

    return total;
}

// Function to calculate the average marks
```

```

function calculateAverage(values) {
    return calculateTotal(values) / values.length;
}

// Function to find the highest mark
function findHighest(values) {
    return Math.max(...values);
}

// Function to find the lowest mark
function findLowest(values) {
    return Math.min(...values);
}

// Function to display the results
function displayResult() {
    console.log("----- Student Marks Analysis -----");
    console.log("Marks:", marks);
    console.log("Total Marks:", calculateTotal(marks));
    console.log("Average Marks:", calculateAverage(marks).toFixed(2));
    console.log("Highest Mark:", findHighest(marks));
    console.log("Lowest Mark:", findLowest(marks));
}

// Calling the function
displayResult();

```

7. CODE EXPLANATION

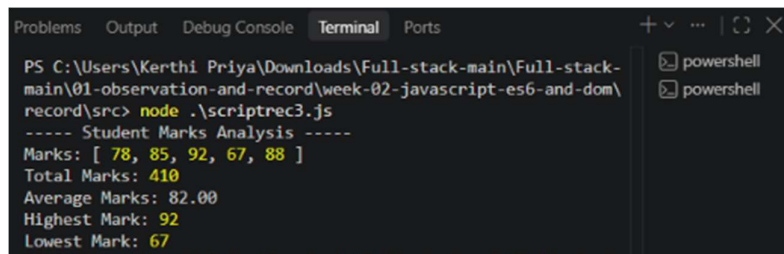
S.No.	Code / Component	Explanation
1	let marks = [78, 85, 92, 67, 88];	Creates an array containing the marks of five students.
2	function calculateTotal(values)	Defines a function to calculate the total marks.
3	let total = 0;	Initializes the total to zero before adding the marks.
4	for (let mark of values)	Uses a for...of loop to access each mark in the array.
5	total += mark;	Adds each mark to the total.
6	function calculateAverage(values)	Defines a function to calculate the average marks.
7	values.length	Returns the number of elements in the marks array.

8	Math.max(...values)	Finds the highest mark by passing all array values to Math.max().
9	Math.min(...values)	Finds the lowest mark using Math.min().
10	function displayResult()	Defines a function to display all calculated results.
11	console.log()	Displays the marks and results in the console.
12	.toFixed(2)	Formats the average to two decimal places.
13	displayResult();	Calls the function and starts the marks analysis.

8. TEST CASES AND EXECUTION DATA

Test Case 1 – Program Execution

Input / Action	Expected Result	Status
Execute the JavaScript program with marks = [78, 85, 92, 67, 88]	Student Marks Analysis and all calculated results are displayed in the console	Pass



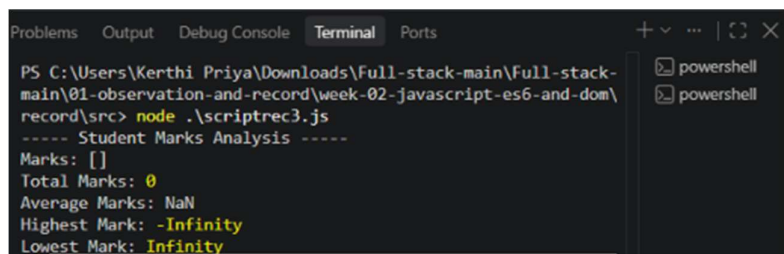
```

PS C:\Users\Kerthi Priya\Downloads\Full-stack-main\Full-stack-main\01-observation-and-record\week-02-javascript-es6-and-dom\record\src> node .\scriptrec3.js
----- Student Marks Analysis -----
Marks: [ 78, 85, 92, 67, 88 ]
Total Marks: 410
Average Marks: 82.00
Highest Mark: 92
Lowest Mark: 67

```

Test Case 2 – Empty Array

Input / Action	Expected Result	Status
Change the marks array to [] and execute the program	Average becomes NaN, highest becomes -Infinity, and lowest becomes Infinity because the program has no empty-array validation	Fail



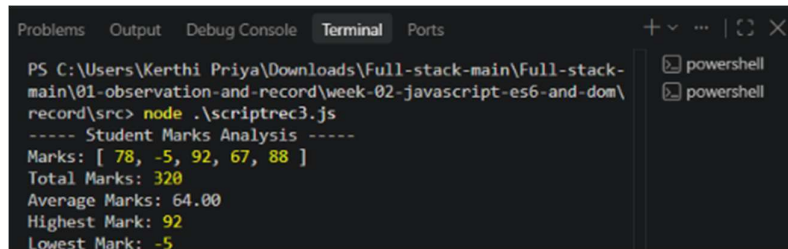
```

PS C:\Users\Kerthi Priya\Downloads\Full-stack-main\Full-stack-main\01-observation-and-record\week-02-javascript-es6-and-dom\record\src> node .\scriptrec3.js
----- Student Marks Analysis -----
Marks: []
Total Marks: 0
Average Marks: NaN
Highest Mark: -Infinity
Lowest Mark: Infinity

```

Test Case 3 – Invalid Negative Mark

Input / Action	Expected Result	Status
Change the marks array to [78, -5, 92, 67, 88]	Program accepts the negative mark because no validation is implemented	Fail



```
PS C:\Users\Kerthi Priya\Downloads\Full-stack-main\Full-stack-main\01-observation-and-record\week-02-javascript-es6-and-dom\record\src> node .\scriptrec3.js
----- Student Marks Analysis -----
Marks: [ 78, -5, 92, 67, 88 ]
Total Marks: 320
Average Marks: 64.00
Highest Mark: 92
Lowest Mark: -5
```

Test Case 4 – Average Marks Calculation

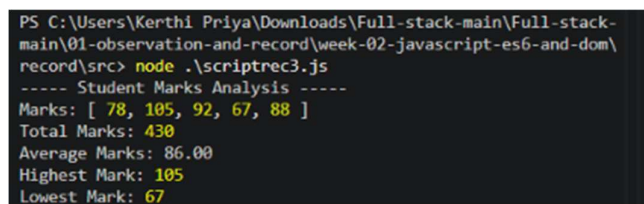
Input / Action	Expected Result	Status
Calculate average for marks 78, 85, 92, 67, 88	Average Marks is displayed as 82.00	Pass



```
Total Marks: 410
Average Marks: 82.00
Highest Mark: 92
```

Test Case 5 – Mark Greater Than 100

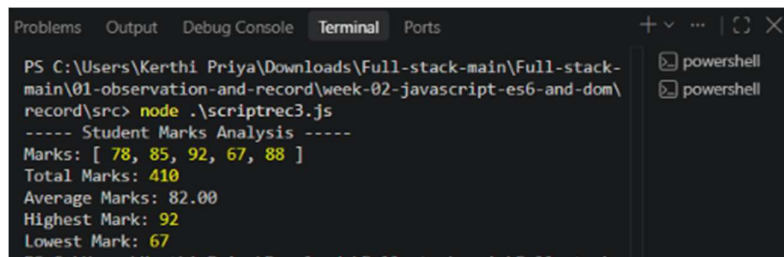
Input / Action	Expected Result	Status
Change the marks array to [78, 105, 92, 67, 88]	Program accepts the mark greater than 100 because no validation is implemented	Fail



```
PS C:\Users\Kerthi Priya\Downloads\Full-stack-main\Full-stack-main\01-observation-and-record\week-02-javascript-es6-and-dom\record\src> node .\scriptrec3.js
----- Student Marks Analysis -----
Marks: [ 78, 105, 92, 67, 88 ]
Total Marks: 430
Average Marks: 86.00
Highest Mark: 105
Lowest Mark: 67
```

9. OUTPUT

Output 1 – Successful Execution



```
PS C:\Users\Kerthi Priya\Downloads\Full-stack-main\Full-stack-main\01-observation-and-record\week-02-javascript-es6-and-dom\record\src> node .\scriptrec3.js
----- Student Marks Analysis -----
Marks: [ 78, 85, 92, 67, 88 ]
Total Marks: 410
Average Marks: 82.00
Highest Mark: 92
Lowest Mark: 67
```

The screenshot shows a terminal window with a dark background. The command prompt is 'PS C:\Users\Kerthi Priya\Downloads\Full-stack-main\Full-stack-main\01-observation-and-record\week-02-javascript-es6-and-dom\record\src>'. The command entered is 'node .\scriptrec3.js'. The output is as follows: '----- Student Marks Analysis -----', 'Marks: [78, 85, 92, 67, 88]', 'Total Marks: 410', 'Average Marks: 82.00', 'Highest Mark: 92', and 'Lowest Mark: 67'. The terminal window has tabs for 'Problems', 'Output', 'Debug Console', 'Terminal', and 'Ports'. The 'Terminal' tab is active. There are also icons for opening a new terminal, saving, and closing the terminal.

10. RESULT

The JavaScript program was successfully implemented using **arrays and user-defined functions**. The program calculates and displays the **total, average, highest, and lowest marks** correctly. For the given marks, the total is **410**, average is **82.00**, highest mark is **92**, and lowest mark is **67**.